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CMSC 471

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QUESTION 1

For this $d(n)$ is the number of positions in the current word that differ from the goal word.

For the steps cost measure, my heuristic is: $h(n) = d(n)$

Each legal action changes exactly one letter and costs 1. Therefore, every position that differs from the goal has to be changed at least once before reaching the goal. If there are $d(n)$ mismatched positions, at least $d(n)$ actions are still required.

Because the heuristic only counts these unavoidable changes, it cannot overestimate the true remaining cost. At the goal, there are no mismatched positions, so the heuristic returns 0. Therefore, this heuristic is admissible.

$h(n) = \text{sum of } V(\text{goal}[i]) \text{ for every position } i \text{ where } \text{state}[i] \neq \text{goal}[i]$
where $V(c)$ is the Scrabble value of letter c .

For every position where the current word differs from the goal, the correct goal letter must eventually be inserted into that position. Since the cost of an action is the Scrabble value of the replacement letter, the cost of inserting each of these goal letters is unavoidable.

The heuristic does not include the cost of any additional intermediate letter changes that may be necessary, so the actual remaining cost can only be equal to or greater than the heuristic. At the goal, there are no mismatched positions, so the heuristic returns 0. Therefore, this heuristic is admissible.

For the frequency cost measure, my heuristic is: $h(n) = d(n) + \text{rarity}(\text{goal})$

for a non-goal state, and $h(\text{goal}) = 0$.

Under the frequency objective, entering a word w costs: $1 + \text{rarity}(w)$ and all rarity values are nonnegative.

If there are $d(n)$ mismatched positions, at least $d(n)$ actions are required because each action changes only one letter. Each of these actions costs at least 1. In addition, the final action must enter the goal word, so the rarity cost of the goal is unavoidable.

Therefore, the minimum possible remaining cost is at least: $(d(n) - 1) + (1 + \text{rarity}(\text{goal}))$

which simplifies to: $d(n) + \text{rarity}(\text{goal})$.

The heuristic ignores the rarity costs of intermediate words, so it cannot overestimate the true remaining cost. It also returns 0 at the goal. Therefore, the frequency heuristic is admissible.

QUESTION 2

1. **Did UCS and A* always return the same optimal cost? If not, identify and investigate the discrepancy.**

Yes, for all six experiments, UCS and A* returned the same optimal cost. For WHY → ASK, the costs were 5 for steps, 9 for Scrabble, and about 38.8554 for frequency. For BEAR → DUCK, the costs were 4 for steps, 11 for Scrabble, and about 38.2402 for frequency. This is consistent with the heuristics being admissible, because A* should still return an optimal solution when using an admissible heuristic.

2. **For each word pair, under which cost measure did A* produce the largest percentage reduction in expanded nodes?**

For both word pairs, the Scrabble heuristic produced the largest percentage reduction in expanded nodes. WHY → ASK had a 98.30% reduction, while BEAR → DUCK had a 99.80% reduction.

3. **For each word pair, under which cost measure was the heuristic least informative?**

The frequency heuristic was the least informative for both word pairs. For WHY → ASK it reduced expansions by only 21.53%, compared with over 97% for the other two cost measures. For BEAR → DUCK it reduced expansions by 72.14%, which was still substantially less than the reductions from the steps and Scrabble heuristics.

4. **Explain the differences using the definitions of your heuristics, not runtime alone.**

The differences in node expansion are caused by how much information each heuristic captures about the corresponding path cost. The steps heuristic counts the number of mismatched character positions, which directly lower-bounds the minimum number of remaining actions. The Scrabble heuristic is more informative because it adds the Scrabble values of the goal letters that must eventually be inserted at mismatched positions, giving a stronger lower bound on the leftover Scrabble cost. The frequency heuristic is weaker because it uses the number of required changes plus the rarity of the goal word, but it cannot reliably predict the rarity costs of intermediate words. As a result, many states receive similar or relatively low heuristic values, so A* must explore more of the search space.

Word Pair	Cost Measure	UCS Cost	UCS Expanded	A* Cost	A* Expanded	Reduction
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WHY → ASK	Steps	5	773	5	16	97.93%
WHY → ASK	Scrabble	9	765	9	13	98.30%
WHY → ASK	Frequency	38.8554	808	38.8554	634	21.53%
BEAR → DUCK	Steps	4	691	4	6	99.13%
BEAR → DUCK	Scrabble	11	2981	11	6	99.80%
BEAR → DUCK	Frequency	38.2402	1084	38.2402	302	72.14%

QUESTION 3

For the steps cost measure, the optimal cost from W1 to W2 will always equal the optimal cost from W2 to W1. Every legal action costs 1, so any path has the same cost when it is followed in reverse. If the shortest path from one word to another has k steps, the reverse path also has k steps, so the optimal costs are equal. For the Scrabble cost measure, the answer is no because the cost depends on the replacement letter being inserted. For example, dog and cog are both legal dictionary words and differ by one letter. Going from dog to cog inserts c, which costs 3, while going from cog to dog inserts d, which costs 2. In this case those are also the optimal costs, so the two directions are different. For the frequency cost measure, the answer is also no because entering a word w costs $1 + \text{rarity}(w)$. In the provided dictionary, dog has rarity 5.819885 and cog has rarity 11.393166. Therefore dog to cog costs $1 + 11.393166 = 12.3931661$, while cog to dog costs $1 + 5.819885 = 6.8198851$, and these are the optimal one-step costs for those two directions. This shows that DOGCAT actions are reversible, but their costs are only symmetric with the steps measure. Scrabble costs depend on which letter is inserted, and frequency costs depend on which word is entered.

QUESTION 4

I used ChatGPT 5.6-Sol for this question. The exact prompt I gave it was the prompt provided in the assignment:

“In DOGCAT, the initial and goal states are lowercase words of the same length, three or four letters and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary. The baseline heuristic $h_0(n)$ is the number

of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal. By ‘more informative,’ we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.”

The model said to use the heuristic: $h(n) = h_0(n) + \text{rarity}(\text{goal})$

for every non-goal state, and $h(\text{goal}) = 0$

Here, $h_0(n)$ is the number of positions where the current word differs from the goal. The idea is that each mismatched position must still be changed at least once, so at least $h_0(n)$ actions remain. Under the frequency objective, every action costs at least 1 because rarity values are nonnegative. Also, any solution from a non goal state must eventually enter the goal word, and the final action includes the unavoidable cost $\text{rarity}(\text{goal})$. The unit-cost portion of that final action is already included in $h_0(n)$, so adding only $\text{rarity}(\text{goal})$ does not count it twice. Therefore, the heuristic cannot overestimate the true optimal remaining cost and is admissible. The frequency objective itself is defined as $1 + \text{rarity}(w)$ for each resulting word.

The model's suggested heuristic is valid, so I didn't need to repair anything. I tested it against the baseline heuristic h_0 on the two frequency cost problems required by the assignment. For $\text{WHY} \rightarrow \text{ASK}$, the baseline heuristic returned an optimal cost of 38.8554 and expanded 755 nodes. The rarity aware heuristic returned the same optimal cost of 38.8554 but expanded only 634 nodes. This is a reduction of about 16.03%. For $\text{BEAR} \rightarrow \text{DUCK}$, the baseline heuristic returned an optimal cost of 38.2402 and expanded 720 nodes. The rarity-aware heuristic returned the same optimal cost of 38.2402 while expanding only 302 nodes, a reduction of approximately 58.06%.

These experiments support using the rarity aware heuristic instead of h_0 for these frequency cost problems. It kept the same optimal solution cost in both cases while reducing the number of nodes expanded. The improvement was much larger for $\text{BEAR} \rightarrow \text{DUCK}$ than for $\text{WHY} \rightarrow \text{ASK}$, so the amount of benefit depends on the particular search problem. These experiments show that the heuristic is more useful in practice, while the admissibility argument above is what shows that it does not overestimate the remaining cost.

AI USE DISCLOSURE: I use AI to understand some of the assignment requirements and check some of my code, mostly the looping sections to make sure my looping syntax was correct since I haven't used python in awhile. I also checked the heuristic reasoning against the cost definitions and admissibility requirements in the assignment.